

u-DFOV: User-Activated Dynamic Field of View Restriction for Managing Cybersickness and Task Performance

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ABSTRACT

Dynamic field of view restriction is one effective way of mitigating cybersickness by modulating the amount of visual information during virtual navigation. However, in the presence of an interactive task for which visibility is important, it can impede the task performance. This poster examined the efficiency of users, manually engaging the dynamic field of view restriction to control and mitigate cybersickness while performing interactive tasks. The comparative experiment has shown that the user-activated method significantly reduced cybersickness as much as the automatic method. However, it also achieved significantly higher task performance and usability despite the manual control.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Virtual Reality (VR) is employed across various domains. Nevertheless, despite its widespread usage, the obstacle of cybersickness poses a barrier to reaching its full potential. Cybersickness symptoms typically include nausea, disorientation, sweating, and eye fatigue [1–3]. These symptoms not only contribute to an unpleasant experience for users but also hinder the prolonged use of VR.

Several strategies have been proposed to address this issue. Seay et al. [3] tackled this issue by reducing the users' field of view (FOV). Their findings revealed significant differences in the extent of the cybersickness - namely higher when participants were exposed to a larger FOV. Building upon these findings, Fernandes et al. [1] developed a dynamic size-shifting FOV respective to the speed and angular velocity of the user. The study compared the Dynamic Field Of View (DFOV) and the controlled conditions (full FOV) and observed a decrease in the average SSQ score. Furthermore, no significant disparity in presence was identified.

Dynamic FOV adjustment/restriction is usually applied automatically according to the navigation motion profile (referred to as the a-DFOV). As humans exhibit a large variance in the sensitivity to cybersickness, an automatic algorithm may not be as effective as for an individual. Moreover, in the presence of an interactive task for which visibility is important, it can impede the task performance.

Yet, known for its effectiveness in mitigating cybersickness, DFOV has been successfully integrated into VR systems. As an illustration, an asset, named "VR Tunnelling Pro," offers users various versions of the DFOV with different features. Furthermore, VR games such as "Population: ONE", "CloudBound" and "Borderlands 2 VR" provide an implementation of DFOV to prevent cybersickness.

Within these virtual environments, users possess the capability to activate or deactivate DFOV before or during the experience. Nevertheless, despite the functionality, it is not clear whether allowing

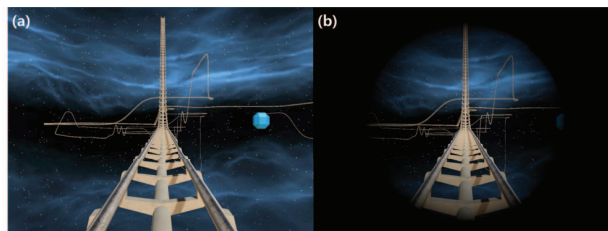


Figure 1: Dynamic FOV to mitigate cybersickness: (a) scene in full FOV and (b) when adjusted with a smaller FOV. Note that the interaction object (blue jewelry) may not be contained in the visual field by the reduced FOV in (b). The experiment compares three conditions: (1) None (full FOV as in (a)), (2) a-DFOV where the FOV is adjusted automatically by the navigation profile (e.g., between (a) and (b)), and (3) u-DFOV in which the automatic DFOV is activated or switched off by the user. When switched on, u-DFOV is equivalent to a-DFOV, and when switched off, it is equivalent to None.

users to discern moments of cybersickness in real-time and switch DFOV on or off, effectively contributes to its mitigation.

As such, in this poster, we conducted a preliminary study of the user-based DFOV restriction (u-DFOV), an approach that allows the user to switch the DFOV feature on and off at their discretion by simply pressing the controller's button. The study's primary goal is to investigate its effectiveness in cybersickness reduction in the sense of catering to the sensitivity of a particular individual. Furthermore, we also take a look at the effect on task performance, usability, and presence/immersion by considering a scenario of virtual navigation (VR roller coaster) with an interactive task (visual search). A human experiment with 18 participants comparing three conditions, namely (1) None (as-is), (2) a-DFOV, and (3) u-DFOV, was conducted.

2 PRELIMINARY VALIDATION EXPERIMENT

The experiment followed 1×3 within-subject design with repeated measures, encompassing three distinct conditions provided in a counter-balanced order. The conditions included: (1) None/Baseline, where no mitigation method was applied, and participants experienced the VR roller coaster with the full FOV (None); (2) automatic DFOV restriction, where the VR content FOV was restricted/adjusted automatically by the navigation speed and angular velocity (a-DFOV); and (3) user-based DFOV restriction, where participants could switch between the dynamic FOV and the full FOV of the VR content using the controller (u-DFOV).

The roller coaster ride, viewed through a VIVE Pro eye head-mounted display, lasted 270 seconds. It was designed to induce a sufficient amount of cybersickness and was implemented using the Unity 3D game engine. While riding the roller coaster, participants were asked to perform interactive tasks. Every 20 seconds, a jewelry object appeared with an audio cue somewhere in the front area of the user as shown in Figure 1. Participants had to quickly locate and select the jewelry using the controller within a 5-second time frame. In each treatment, such objects popped up 12 times.

18 participants (8 males, 10 females, age 19 – 35, $M = 25$, SD

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= 4.03) were recruited from the local communities. During the recruitment process, individuals with high susceptibility to cybersickness were excluded based on Motion Sickness Susceptibility Questionnaire (MSSQ) score (> 35). The average MSSQ score among recruited participants was 16.71 ($SD = 7.11$). This study was approved by the University's Institutional Review Board (No. KUIRB-2023-0296-01). Upon the completion of the experiment, participants received a monetary compensation of \$10.

The task completion time was measured, and the Simulator Sickness Questionnaire (SSQ) [2] was used to measure the level of cybersickness. In addition, after the whole experimental session was over, participants were asked to rank their preferences for usability/comfort and presence/immersion. We omit further details due to lack of space.

3 RESULTS AND DISCUSSION

3.1 Cybersickness Mitigation Effect

The repeated measure ANOVA showed significant differences in the Total SSQ ($F = 5.79$; $p = .006^{**}$). The post-hoc with Bonferroni correction revealed that there were two significant pairs: (a-DFOV $>$ None; $p = .006^{**}$) and (u-DFOV $>$ None; $p = .016^{*}$) (refer to Figure 2). Category-wise, significant differences were found: for Nausea ($F = 6.74$; $p = .003^{**}$) with a-DFOV ($p = .003^{**}$) and u-DFOV ($p = .008^{**}$) showing lower levels than None; for Oculomotor ($F = 4.09$; $p = .025^{*}$) with a-DFOV reporting a lower level than None ($p = .029^{*}$); and for Disorientation ($F = 4.68$; $p = .016^{*}$) with both a-DFOV ($p = .015^{*}$) and u-DFOV ($p = .04^{*}$) showing lower levels than None condition.

Overall, the a-DFOV showed the lowest level of cybersickness compared to the others, followed by the u-DFOV and None. However, there was no statistical difference between the a-DFOV and the u-DFOV. This result suggests that the u-DFOV had the sickness mitigating effect competitive to that of the a-DFOV. This was a bit unexpected, as cybersickness can be typically a progressive symptom rather than an immediate onset. As a result, with the u-DFOV, the user may not have been fast enough to react and switch the DFOV feature on or off to manage one's sickness.

In addition, there could have been situations where the DFOV was turned off to focus on the task and tolerate the sickness for the moment. Despite such a possibility, on the other hand, manual control allows for a personalized prescription for sickness.

In this vein, the activation time of the DFOV feature with the u-DFOV and the relationship to the cybersickness was also analyzed. Participants, on average, engaged the DFOV feature "on" for 150 seconds ($SD = 67.93$) out of the entire 270-second duration of the VR roller coaster content.

Pearson's correlation test showed that there was close to a significant negative correlation between the activation time and the Nausea score ($r = -0.397$; $p = .051$). Disorientation score ($r = -0.226$; $p = .184$), Oculomotor score ($r = -0.253$; $p = .156$), and the Total SSQ Score ($r = -0.303$; $p = .111$) categories also showed negative correlations. This partly suggests that the willful manual control, despite the possible aforementioned delay in its application, can still bring about sickness reduction, especially for Nausea.

3.2 Task Performance

As expected, the u-DFOV showed a clearer edge in the interactive visual task performance than the a-DFOV. The None condition showed the fastest task completion time, as a full FOV is always provided. Significant differences were observed ($F = 66.54$; $p < .001^{***}$) as shown in Figure 3 (a). The post-hoc found three significantly different pairs: a-DFOV $>$ None ($t = 11.42$; $p < .001^{***}$), u-DFOV $>$ None ($t = 4.32$; $p < .001^{***}$), and a-DFOV $>$ u-DFOV ($t = 7.09$; $p < .001^{***}$). The u-DFOV, thus, is an attractive compromise offering efficient task performance with reduced cybersickness.

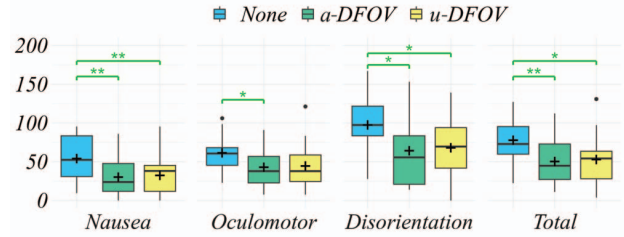


Figure 2: The categorical SSQ scores among the three tested conditions (* $p < .05$; ** $p < .01$).

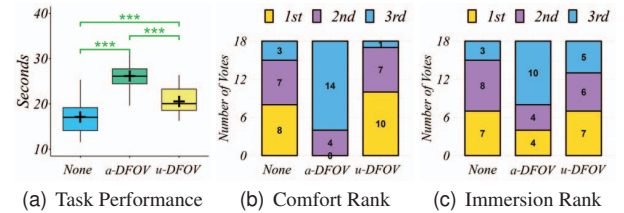


Figure 3: (a) Task completion time (***) and (b, c) Preference ranks for comfort and immersion among the three tested conditions.

3.3 Preference (Comfort and Immersion)

The subject's preference ranks for comfort and immersion, analyzed using the Friedman test, is shown in Figure 3. While the immersion did not show a significant effect ($\chi^2 = 3.11$; $p = .21$), the comfort rank showed a statistically significant difference ($\chi^2 = 16.78$; $p < .001^{***}$). The post-hoc showed that None ($p = .003^{**}$) and u-DFOV ($p < .001^{***}$) were more comfortable than a-DFOV, respectively. The data roughly shows that, despite its effectiveness in reducing sickness, the a-DFOV was the least comfortable or usable due to its interference with task performance. Also, the a-DFOV received the lowest ranking for immersion due to frequent changes in the FOV.

4 CONCLUSION

This poster outlines an empirical study investigating the effect of enabling users to switch their DFOV on and off in real-time in response to their perceived cybersickness level while carrying out a visual task. The experimental results indicate that the u-DFOV showed a significant effect in sickness reduction compared to the None condition despite having to manually manage the sickness level and task achievement. Furthermore, the u-DFOV showed better task performance and user-perceived usability and immersion than the a-DFOV. The u-DFOV, thus, is an attractive way of offering efficient task performance with reduced cybersickness. We speculate that this stems from the "personalized" FOV treatment by the user's own initiation. We plan to further investigate this issue.

ACKNOWLEDGMENTS

This work was supported by the IITP/ITRC Program (IITP-2022-RS-2022-00156354), KIAT/KEA Competency Dev. Program (N0009999), and NRF Korea BRL Program (2022R1A4A1018869).

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